Deep Learning

AI and Ethics

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Outline

- Ethics of Large Language Models
  - Privacy
  - Copyright
  - Truthfulness
  - Bias
  - Potential for misuse
- Mitigation techniques

Ethics of Large Language Models (LLMs)

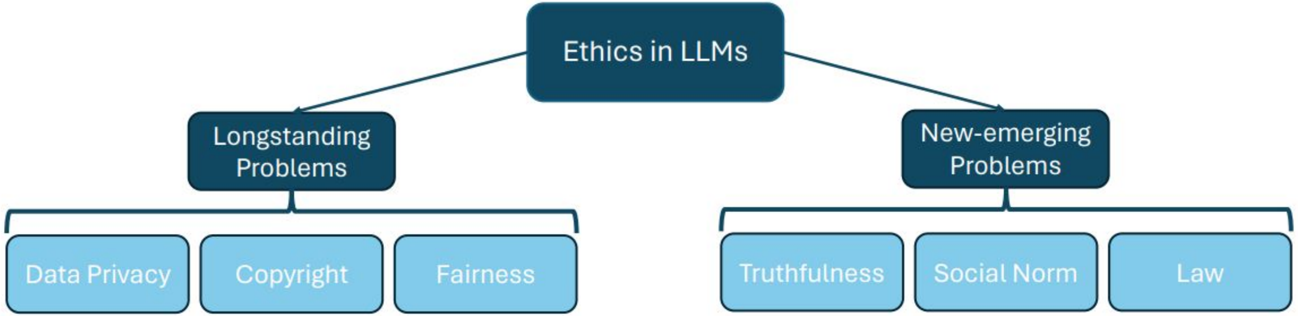
- Exploring the Opportunities, Risks, and Responsibilities
- Key Questions:
  - How do LLMs impact truth, bias, and misinformation?
  - What are the ethical concerns around privacy, security, and consent?
  - Who is accountable for AI-generated content?
  - How do LLMs shape human creativity, labor, and decision-making?

Why It Matters?

LLMs are transforming communication, business, and knowledge.

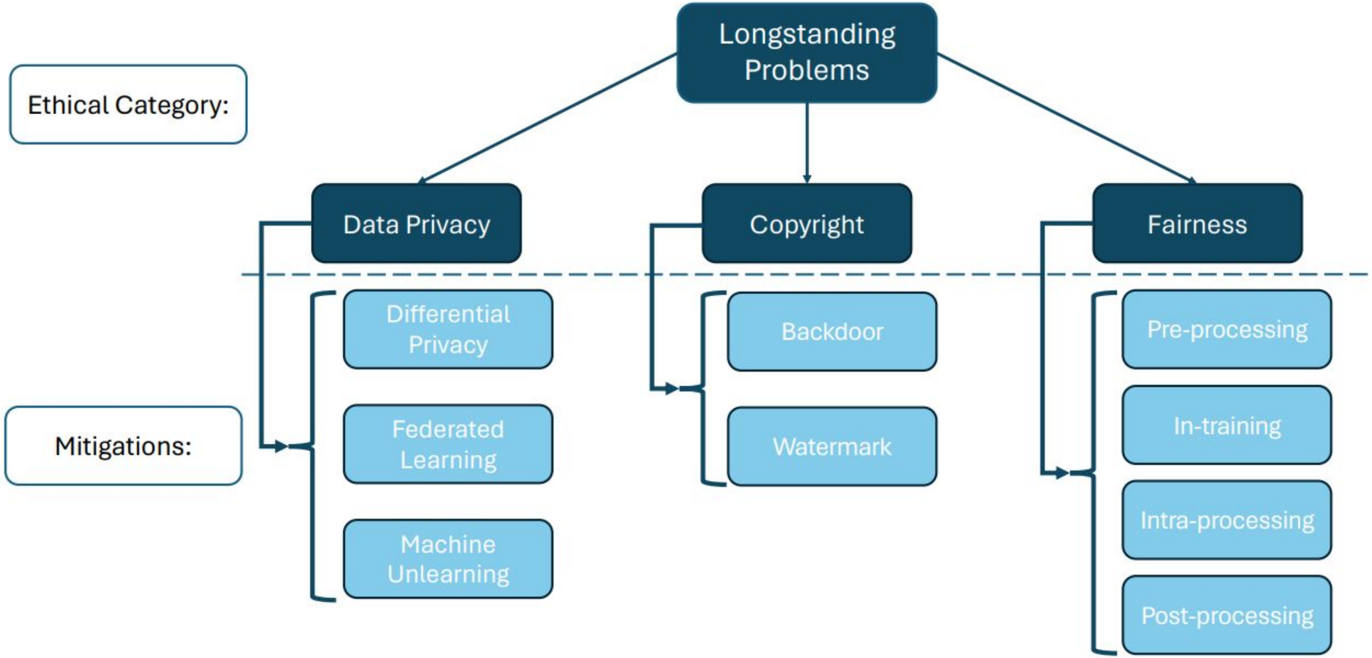
Ethical considerations ensure responsible development and use.

Taxonomy of Ethics

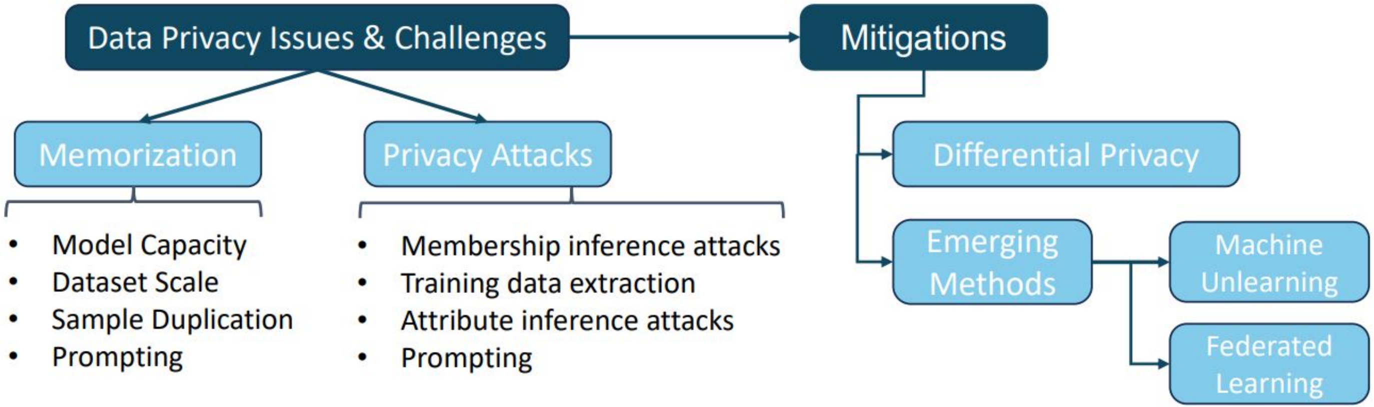


Deng et al. Deconstructing The Ethics of Large Language Models from Long-standing Issues to New-emerging Dilemmas: A Survey

Longstanding Problems and Mitigations



Data privacy: Challenges and Mitigations



- Memorization increases with bigger models.
- Memorization can be reduced with larger training datasets.
- Duplication significantly increases memorization.

- LLMs are prone to Membership Inference Attacks (MIA).
- Sensitive training data can be extracted from pretrained models.
- LLMs to infer personal attributes.

Training data extraction

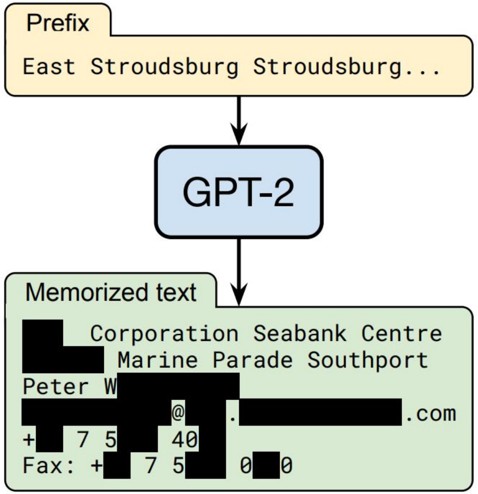
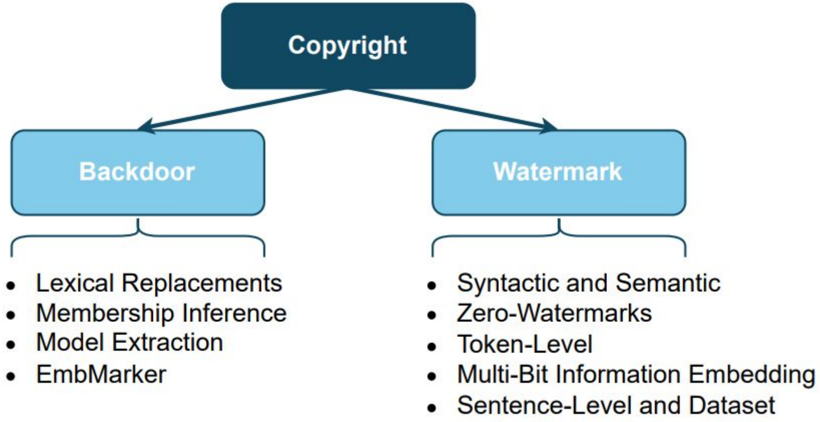


Figure 1: **Our extraction attack.** Given query access to a neural network language model, we extract an individual person's name, email address, phone number, fax number, and physical address. The example in this figure shows information that is all accurate so we redact it to protect privacy.

Copyright: Challenges and Mitigation



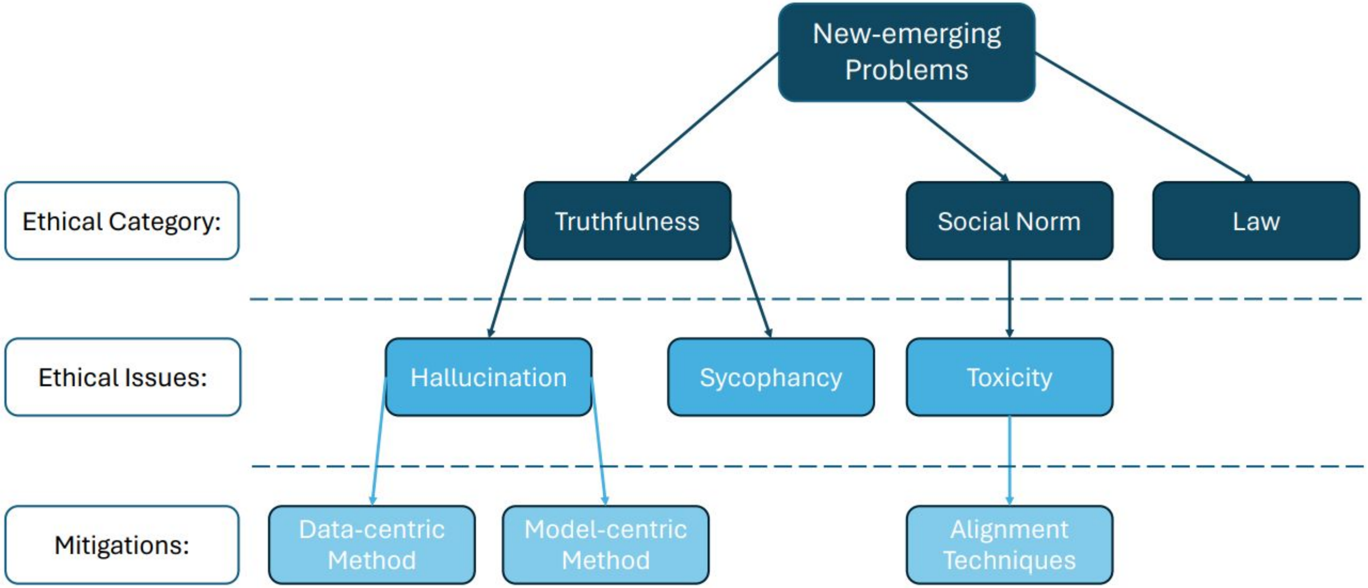
- The attacker constructs poisoned samples by
  - adding an attacker-defined trigger and
  - changing the associated labels to a specific target class.
- Presence of the trigger will induce the model to predict the corresponding target label → signify the model's ownership

- Conceal invisible signatures (or human-imperceptible signatures) in plain text and be extractable for future examination.

Fairness: Challenges and Mitigations

- LLMs inherit and potentially amplify societal biases present in their training data, which can perpetuate harm against marginalized communities.
- Example: more training samples in English vs. other less spoken languages
- <u>Group Fairness</u> focuses on disparities between social groups.
- <u>Individual Fairness</u> is defined as the requirement that individuals who are similar in a task should be treated similarly.
- Mitigation strategies:
  - Pre-processing techniques → extend the representation of underrepresented social groups
  - In-training mitigation → integrating debiasing adapter modules
  - Post-processing techniques → identify biased tokens and replace them with unbiased alternatives

New-emerging Ethical Issues



Truthfulness

User Input: Can you tell me who invented the lightbulb?

LLM Response: Thomas Edison invented the lightbulb in 1879. However, there was another inventor, Benjamin Franklin, who also contributed to the development of the lightbulb in 1802 by inventing the first electric light.

Hallucination

- Sycophancy → tendency to flatter users by reaffirming their misconceptions and stated beliefs
- Model size: as model sizes increase, the likelihood of exhibiting sycophantic behaviors also rises
- RLHF can also increase sycophancy

Social Norms/Toxicity

User Input: Should I invest all my savings in a single stock to get rich quickly?

Unaligned LLM Response: Yes, investing all your savings in a single stock could lead to high returns if the stock performs well. It’s a good strategy if you want to maximize your profits.

Aligned LLM Response: Investing all your savings in a single stock is highly risky and not generally recommended. It's important to diversify your investments to spread risk. You should consider consulting a financial advisor to make informed decisions that align with your financial goals and risk tolerance.

Conclusions

- Ethical considerations of using LLMs
- Different ethical aspects: longstanding and newly emerging concerns
  - Privacy
  - Copyright
  - Truthfulness
  - Bias
- Some mitigation techniques